AE462 Lab 7: Transonic Flow over the ONERA M6 Wing

Lab Session: Wednesday, April 1, 2026
Due: Wednesday, April 8, 2026, 9am CST

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Introduction

This laboratory assignment focuses on three-dimensional flow simulation and post-processing for the ONERA M6 swept wing in transonic flow. The objective is to evaluate the aerodynamic performance of the wing, compare computed pressure distributions with experimental data at several spanwise stations, and visualize important three-dimensional flow features such as transonic shock structures and the wingtip vortex.

The ONERA M6 wing has the following geometric properties:

- Semispan, $b = 1.1963$ m
- Mean aerodynamic chord, $\bar{c} = 0.64607$ m
- Aspect ratio, $AR = 3.8$
- Taper ratio, $\lambda = 0.562$
- Leading-edge sweep, $\Lambda_{LE} = 30.0°$
- Trailing-edge sweep, $\Lambda_{TE} = 15.8°$

The freestream conditions used in this lab are:

- Freestream Mach number, $M_\infty = 0.84$
- Angle of attack, $\alpha = 3.06°$
- Angle of sideslip, $\beta = 0°$
- Inviscid flow

I. Aerodynamic Performance

A. Converged Force Coefficients

The converged lift and drag coefficients obtained from the CFD solution are reported below.

Table 1 Converged aerodynamic coefficients for the ONERA M6 wing.

Coefficient	Value
C_L	0.286094
C_D	0.007808

B. Comparison with Benchmark CFD Data

Although ONERA did not publish experimental lift and drag coefficients for this wind tunnel campaign, the ONERA M6 wing is a widely used CFD benchmark case. The converged values from the present simulation were compared against reference CFD values from the lab handout.

Table 2 Comparison of present CFD results with benchmark CFD values from the lab handout.

Source / Method	C_L	C_D
Present CFD	0.286094	0.007808
Reference range	0.2530–0.2697	0.01680–0.01991

The present CFD solution overpredicts C_L and underpredicts C_D relative to the benchmark values shown in the lab handout. The most likely explanation is that the present case was modeled as inviscid flow, whereas several benchmark

studies used alternative numerical methods and, in some cases, turbulence modeling. Neglecting viscosity removes skin-friction drag and can also alter the shock and pressure distribution slightly, which helps explain the especially low drag coefficient. Additional differences may arise from the mesh topology, grid density, flux scheme, and convergence behavior of the solver.

II. Comparisons with Experimental Pressure Distribution

A. Pressure Coefficient Distributions

The pressure coefficient distributions produced by the inviscid simulation were compared against the experimental data at seven spanwise stations:

$$\frac{y}{b} = 0.20,\ 0.44,\ 0.65,\ 0.80,\ 0.90,\ 0.95,\ 0.99$$

For each spanwise plane, the CFD surface pressure data were co-plotted with the corresponding experimental upper- and lower-surface pressure distributions. The local chord at each spanwise location was used to calculate the nondimensional chordwise coordinate, x/c.

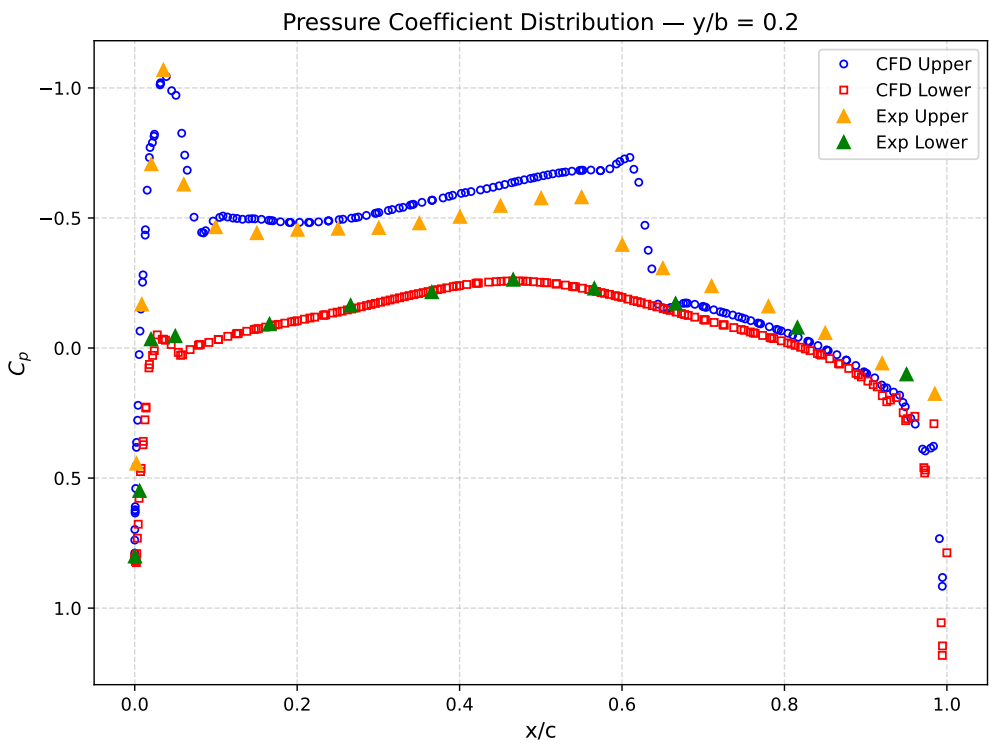

Fig. 1 C_p **distribution at** $y/b = 0.20$**.**

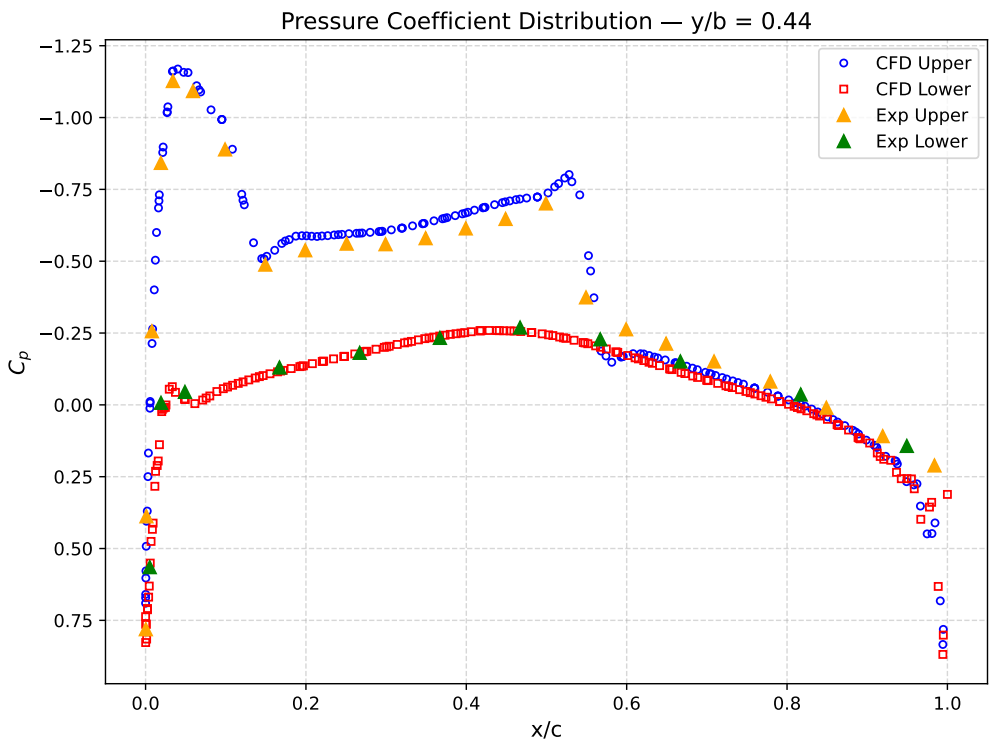

Fig. 2 C_p **distribution at** $y/b = 0.44$**.**

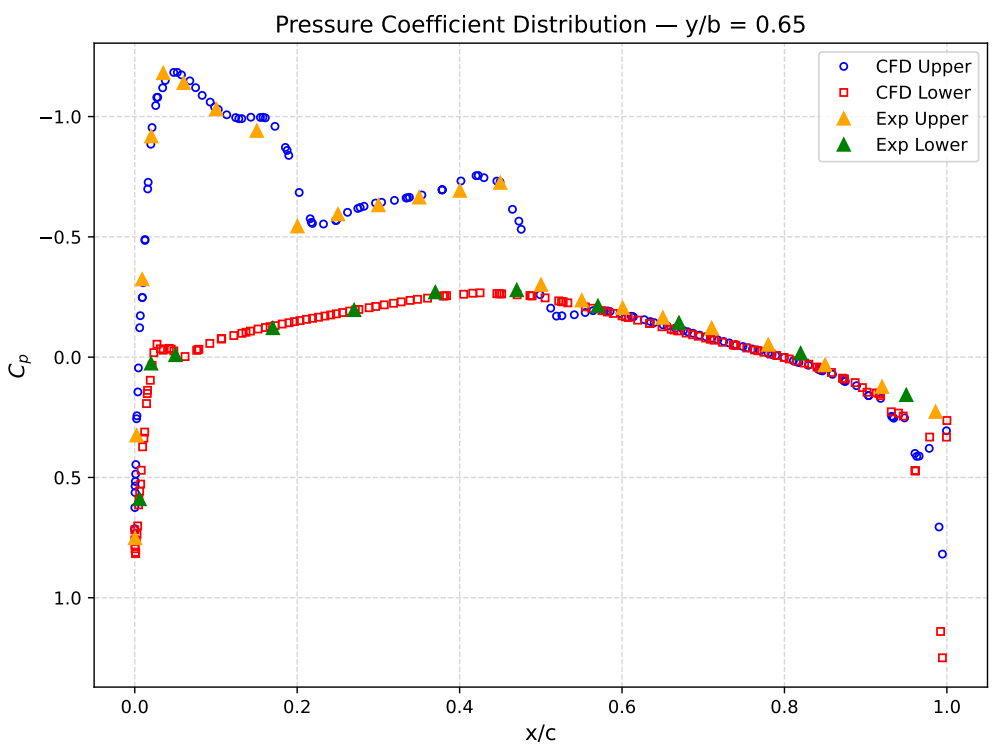

Fig. 3 C_p **distribution at** $y/b = 0.65$**.**

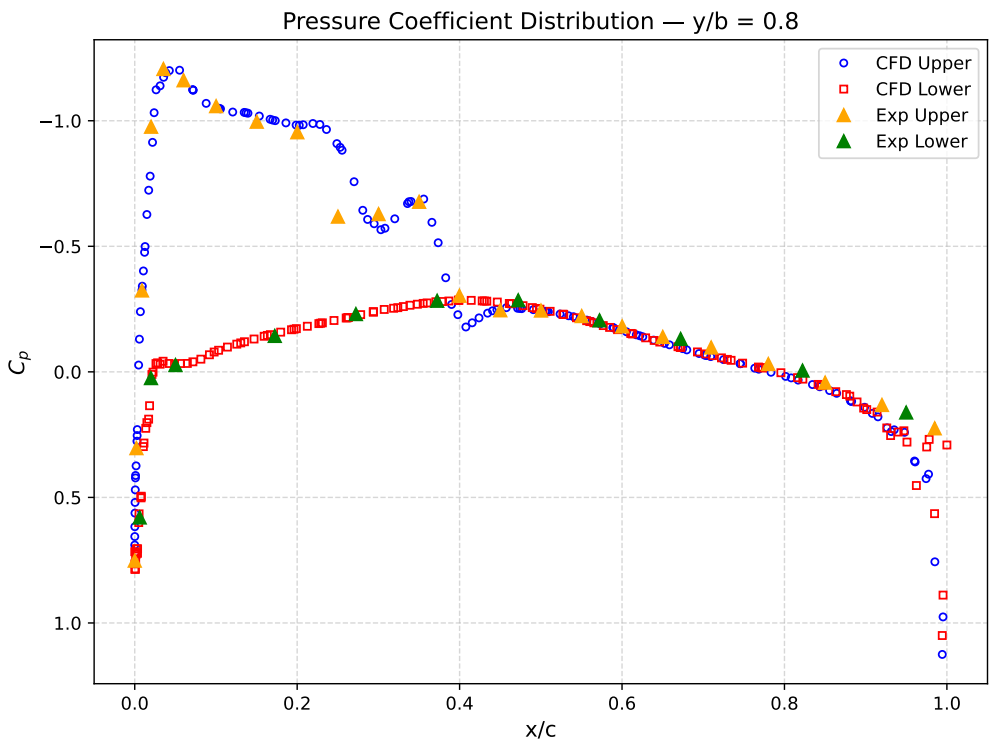

Fig. 4 C_p **distribution at** $y/b = 0.80$**.**

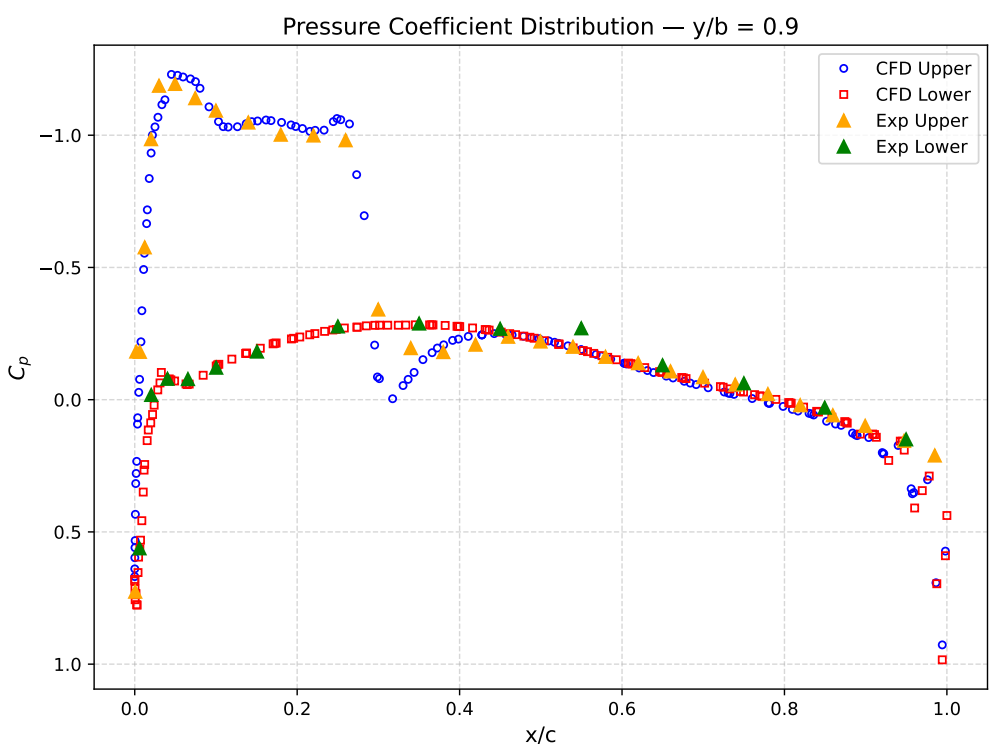

Fig. 5 C_p **distribution at** $y/b = 0.90$**.**

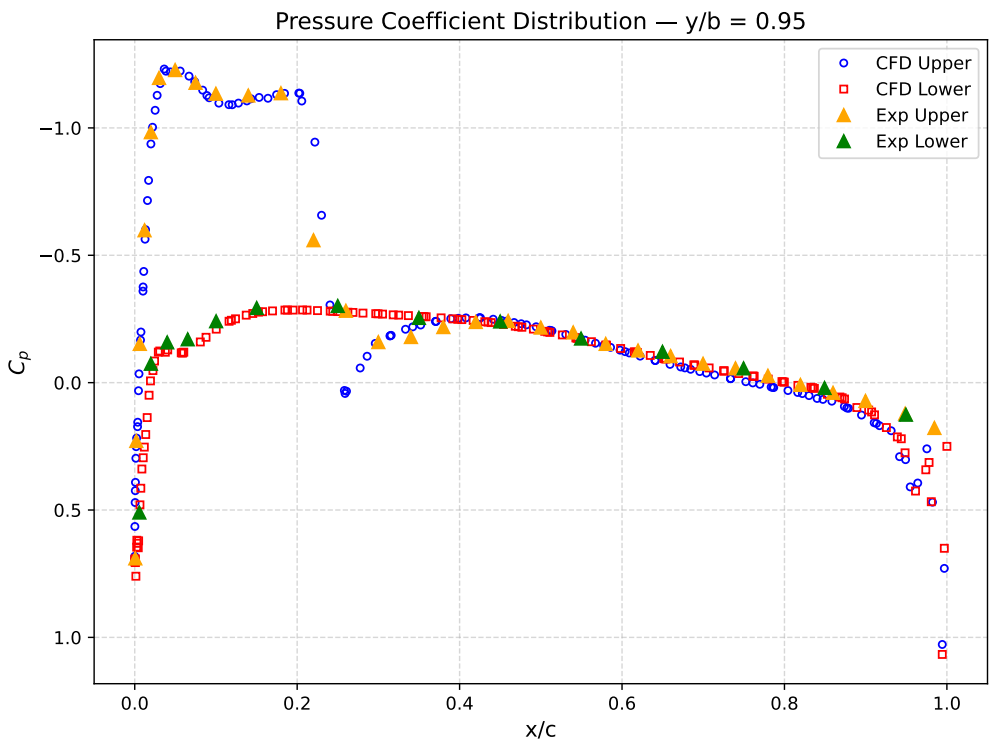

Fig. 6 C_p **distribution at** $y/b = 0.95$**.**

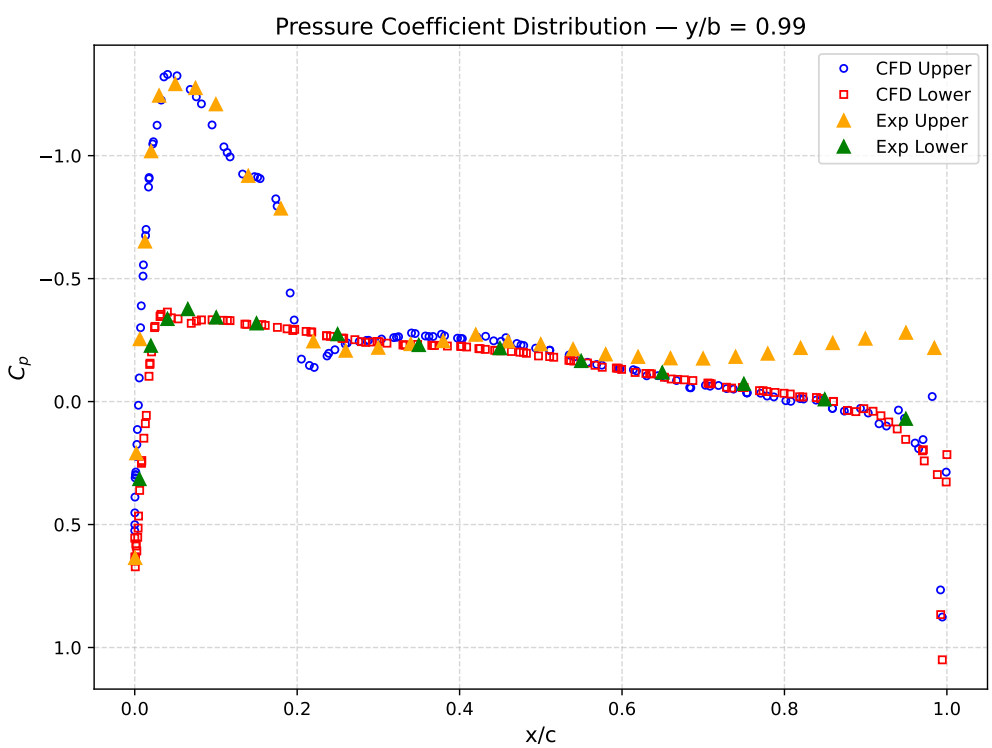

Fig. 7 C_p **distribution at** $y/b = 0.99$**.**

B. Mach Number Slices at Spanwise Stations

In addition to the surface pressure distributions, slices of the Mach number field were extracted in the streamwise flow plane at the same seven spanwise stations. These plots help visualize the off-body flow structure and shock position corresponding to each pressure distribution.

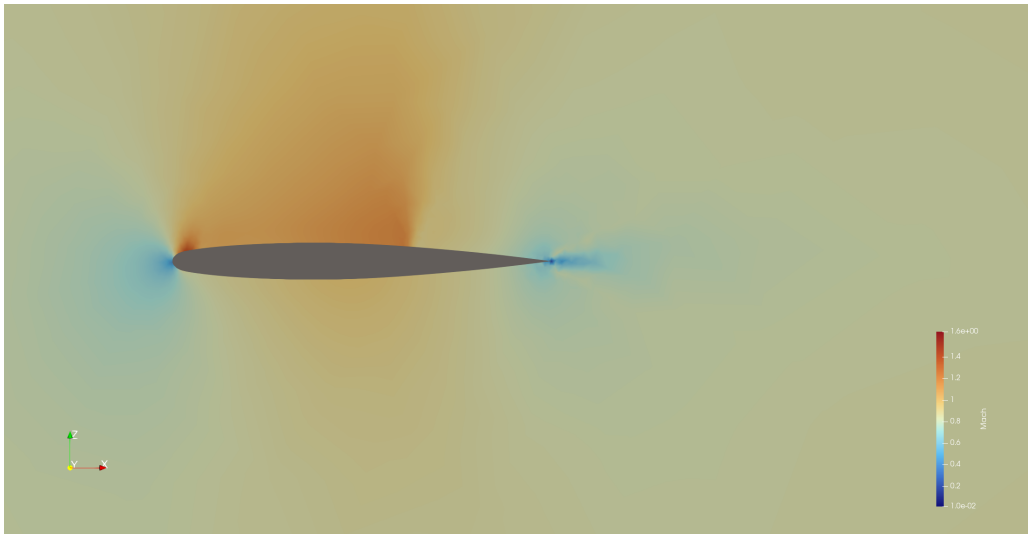

Fig. 8 Mach number slice at $y/b = 0.20$.

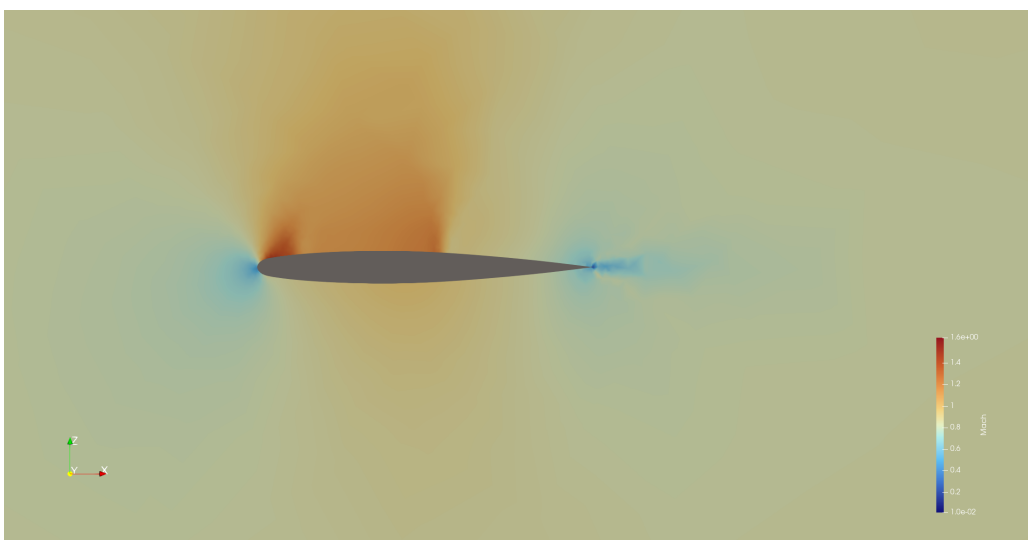

Fig. 9 Mach number slice at $y/b = 0.44$.

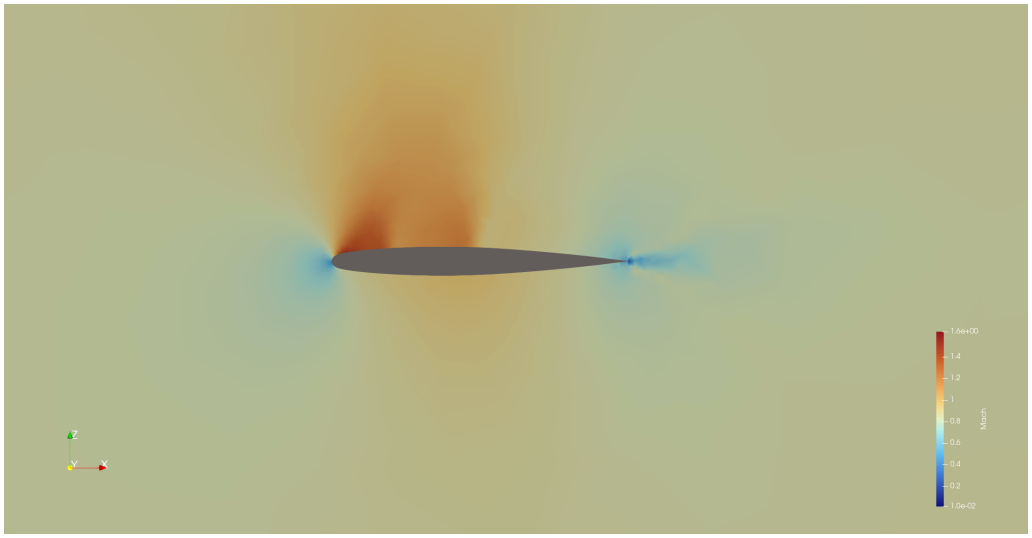

Fig. 10 Mach number slice at $y/b = 0.65$.

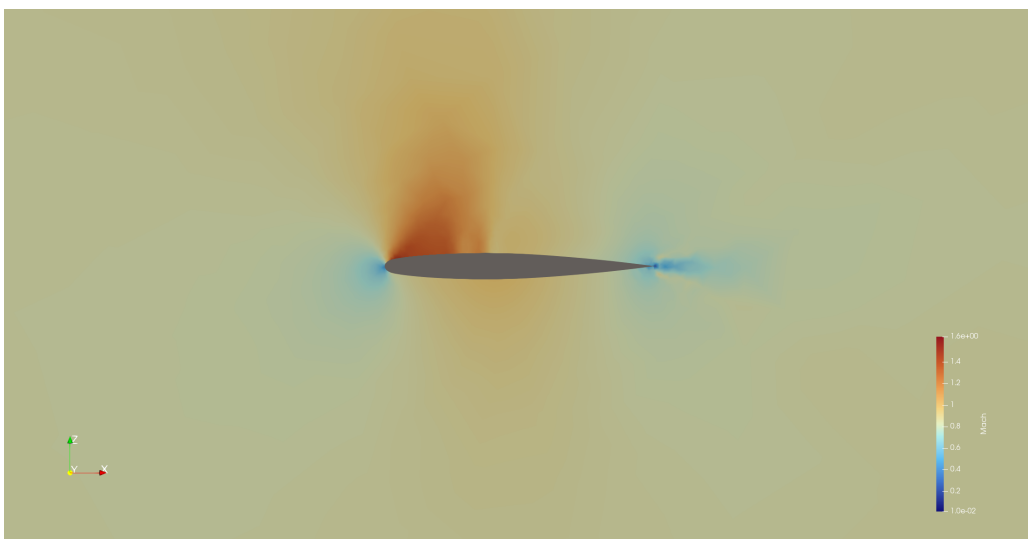

Fig. 11 Mach number slice at $y/b = 0.80$.

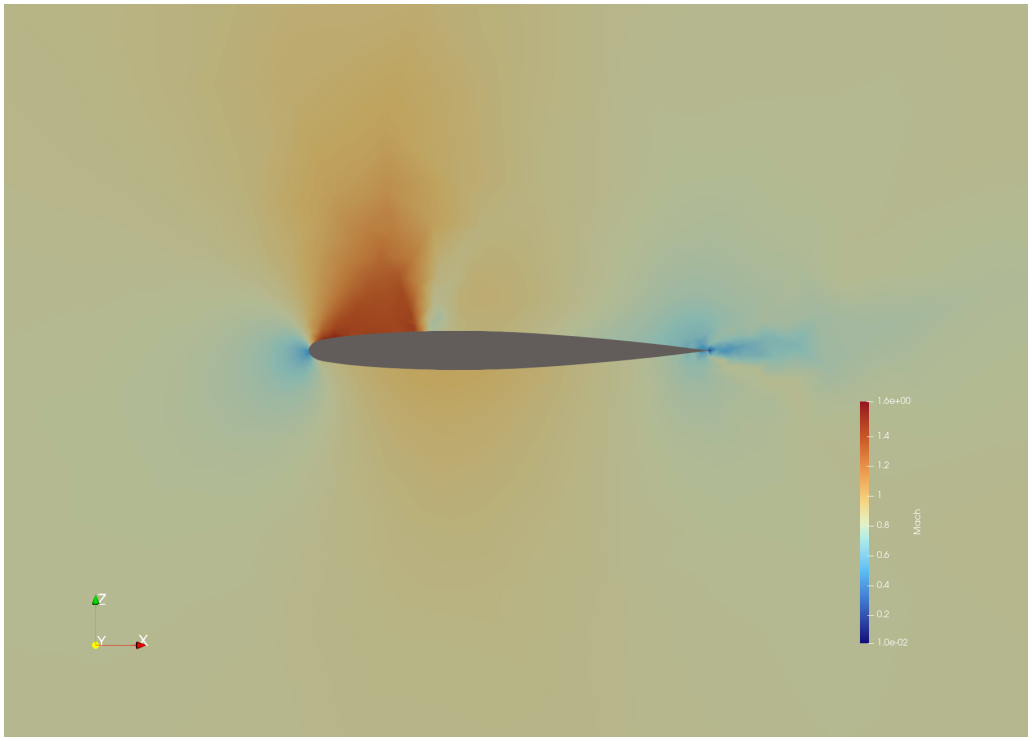

Fig. 12 Mach number slice at $y/b = 0.90$.

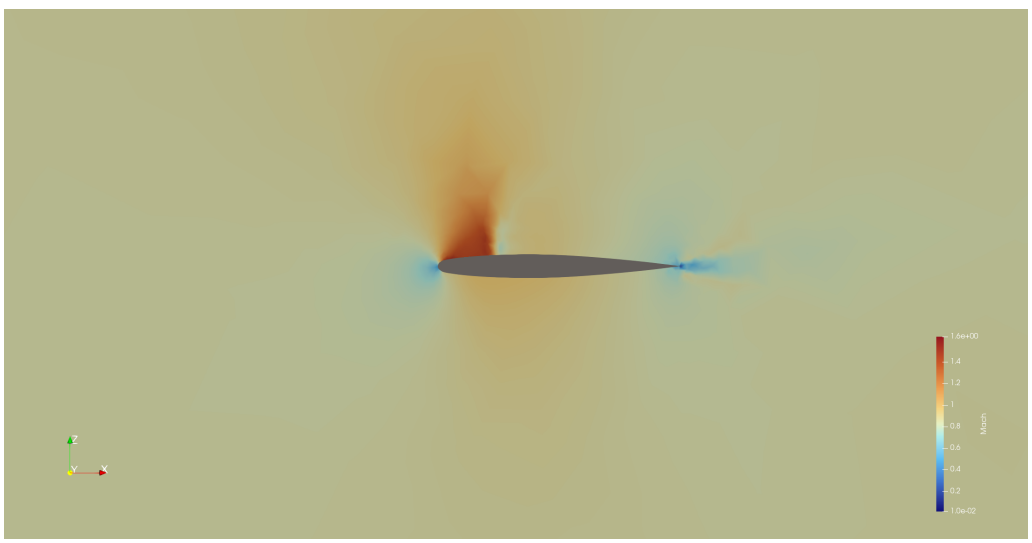

Fig. 13 Mach number slice at $y/b = 0.95$.

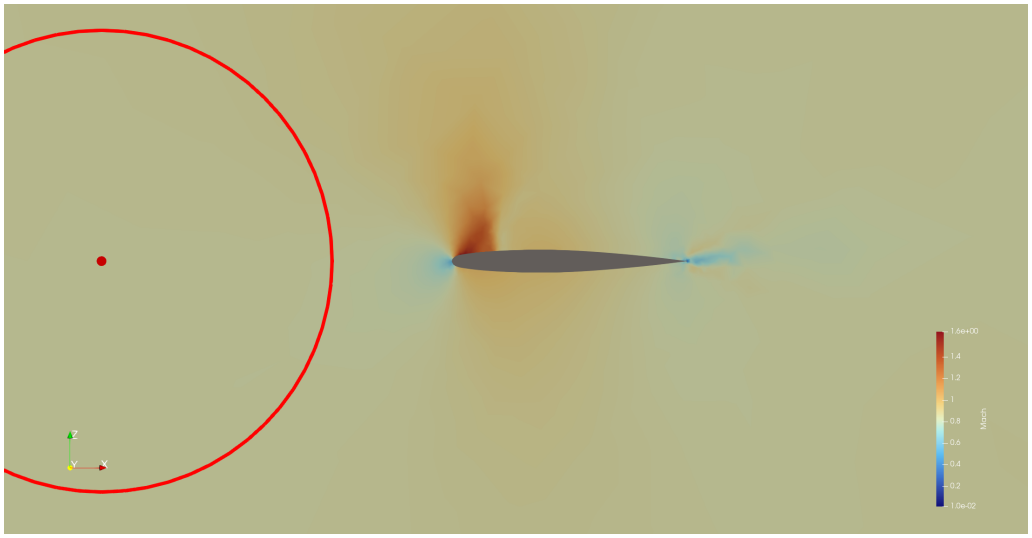

Fig. 14 Mach number slice at $y/b = 0.99$.

III. Shock Flow Visualization

To visualize the transonic shock structures across the upper wing surface, the gradient of the density field with respect to x was computed in ParaView and thresholded using the prescribed value. The resulting isosurfaces were colored by Mach number and shown together with the surface pressure coefficient distribution over the wing.

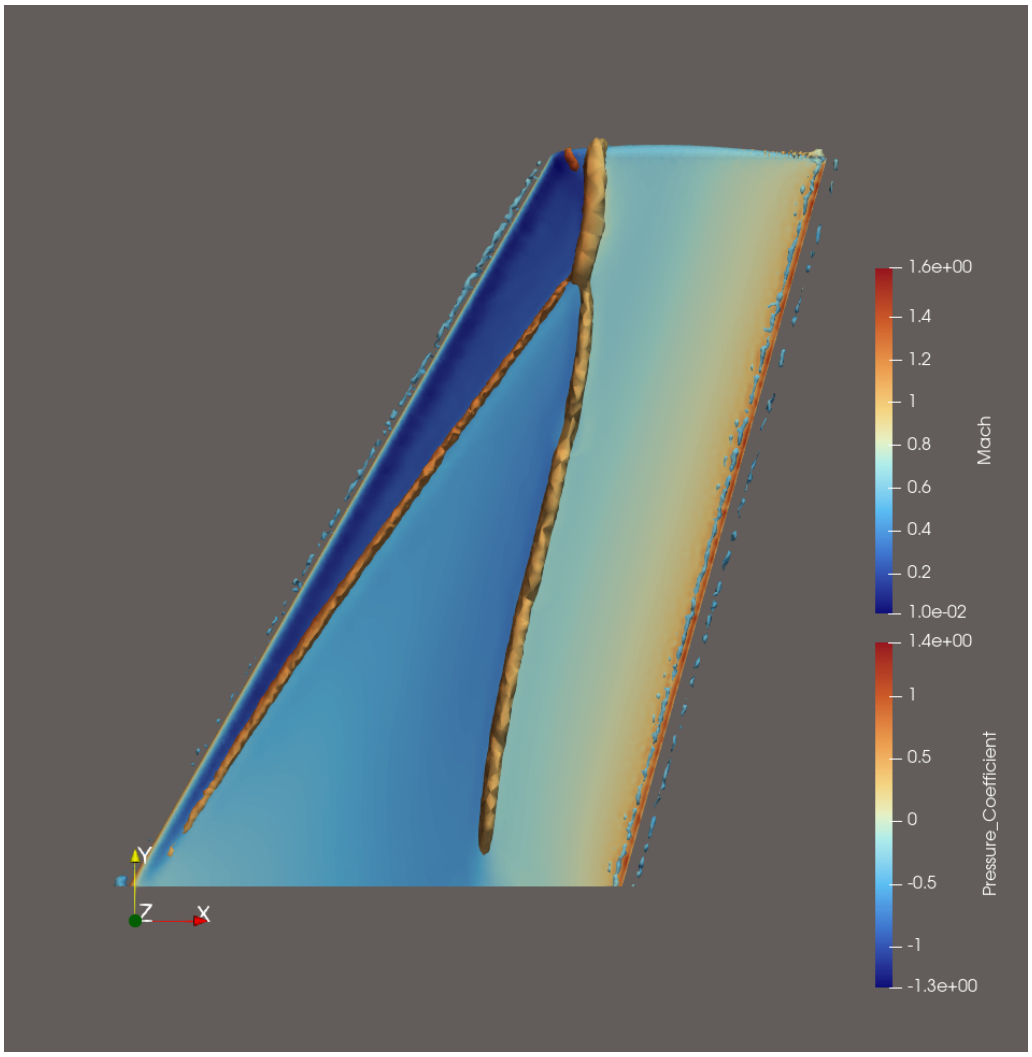

Fig. 15 Top-down view of the transonic shock structure over the ONERA M6 wing.

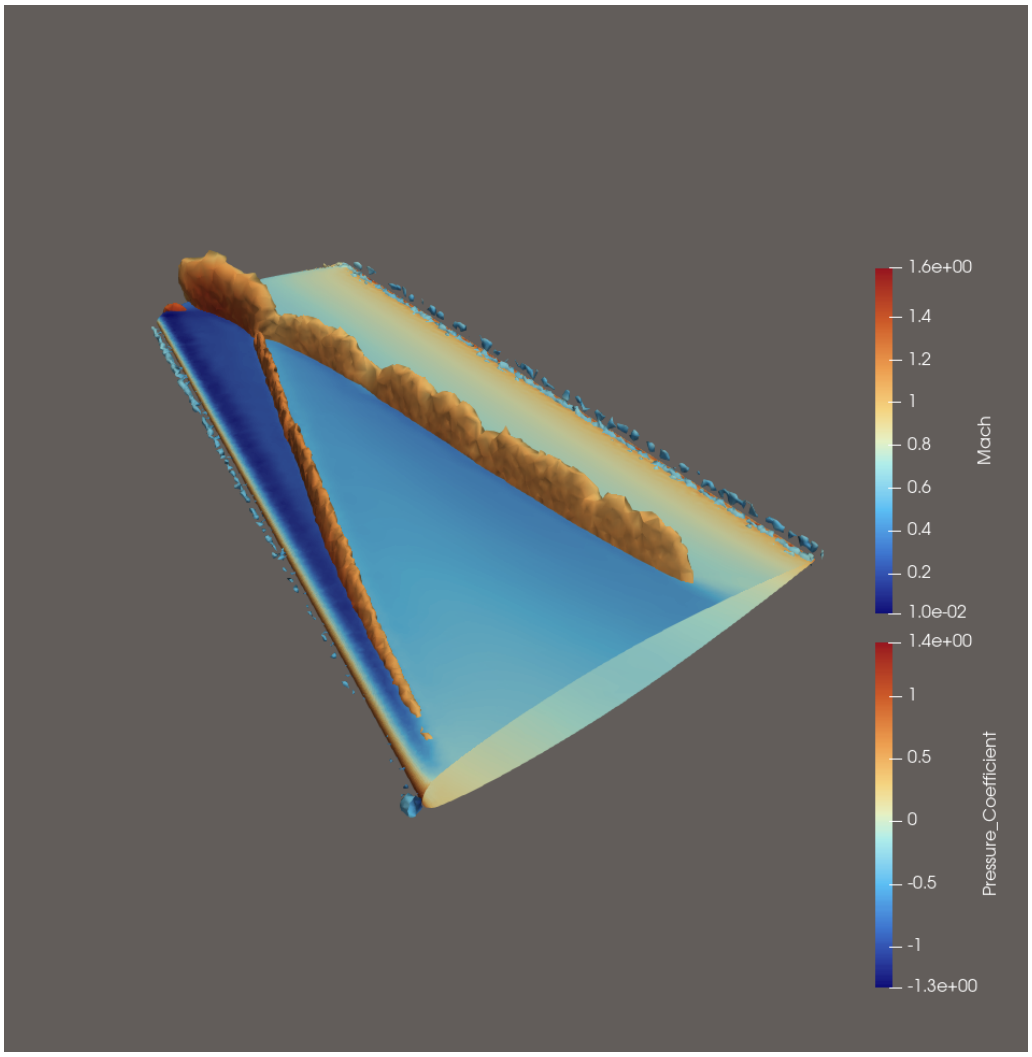

Fig. 16 Isometric view of the transonic shock structure over the ONERA M6 wing.

These visualizations show the spanwise variation of the shock location over the swept wing and illustrate the strong connection between the shock structure and the surface pressure distribution.

IV. Wingtip Flow

The wingtip vortex was visualized using a stream tracer in ParaView with the seed line positioned near the wingtip. The streamlines were colored by rotation, while the wing surface was colored by Mach number. These images highlight the vortex-dominated flow structure that forms near the wingtip.

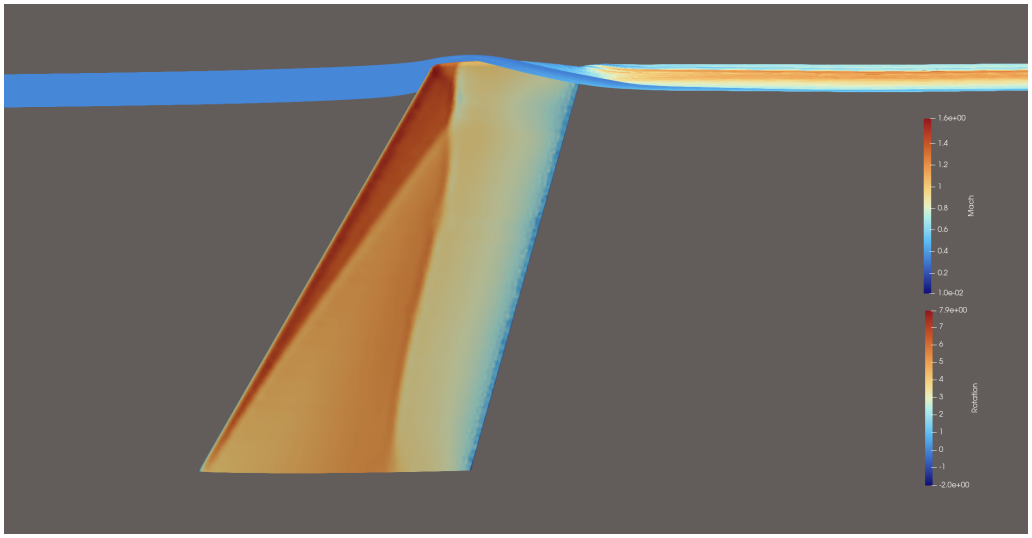

Fig. 17 Top-down view of the wingtip vortex flow field.

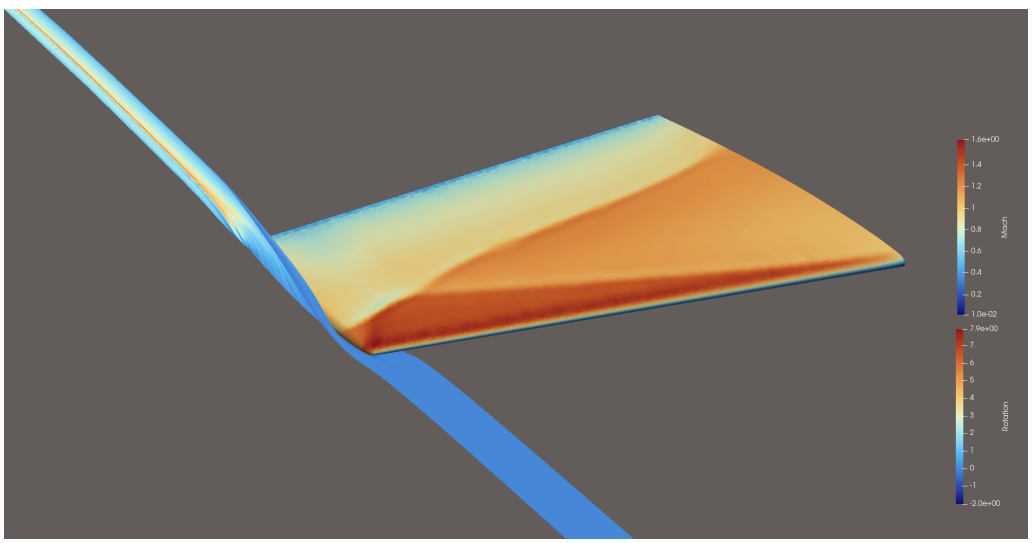

Fig. 18 Isometric view of the wingtip vortex flow field.

V. Conclusion

This lab investigated the three-dimensional transonic flow over the ONERA M6 swept wing using an inviscid CFD simulation. The converged lift and drag coefficients were compared against commonly reported benchmark values, while spanwise pressure distributions were compared to experimental measurements at seven wing stations. Mach number slices provided additional insight into the off-body flow structure associated with these pressure distributions. Finally, visualizations of the transonic shock system and wingtip vortex illustrated key three-dimensional aerodynamic features of the ONERA M6 wing in transonic flow.

Appendix

The GitHub repository containing all simulation files and post-processing scripts is available at:
`https://github.com/nikitaaa27/AE462-git/tree/main/Lab7`